

H3 Definite integrals

Name: _____

Class: _____

Date: _____

Time: **427 minutes**

Marks: **367 marks**

Comments:

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Q1.

- (a) Find the first three terms, in ascending powers of x , of the binomial expansion of

$$\frac{1}{\sqrt{4+x}}$$

(3)

- (b) Hence, find the first three terms of the binomial expansion of $\frac{1}{\sqrt{4-x^3}}$

(2)

- (c) Using your answer to part (b), find an approximation for $\int_0^1 \frac{1}{\sqrt{4-x^3}} dx$, giving your answer to seven decimal places.

(3)

- (d) (i) Edward, a student, decides to use this method to find a more accurate value for the integral by increasing the number of terms of the binomial expansion used.

Explain clearly whether Edward's approximation will be an overestimate, an underestimate, or if it is impossible to tell.

(2)

- (ii) Edward goes on to use the expansion from part (b) to find an approximation

for $\int_{-2}^0 \frac{1}{\sqrt{4-x^3}} dx$

Explain why Edward's approximation is invalid.

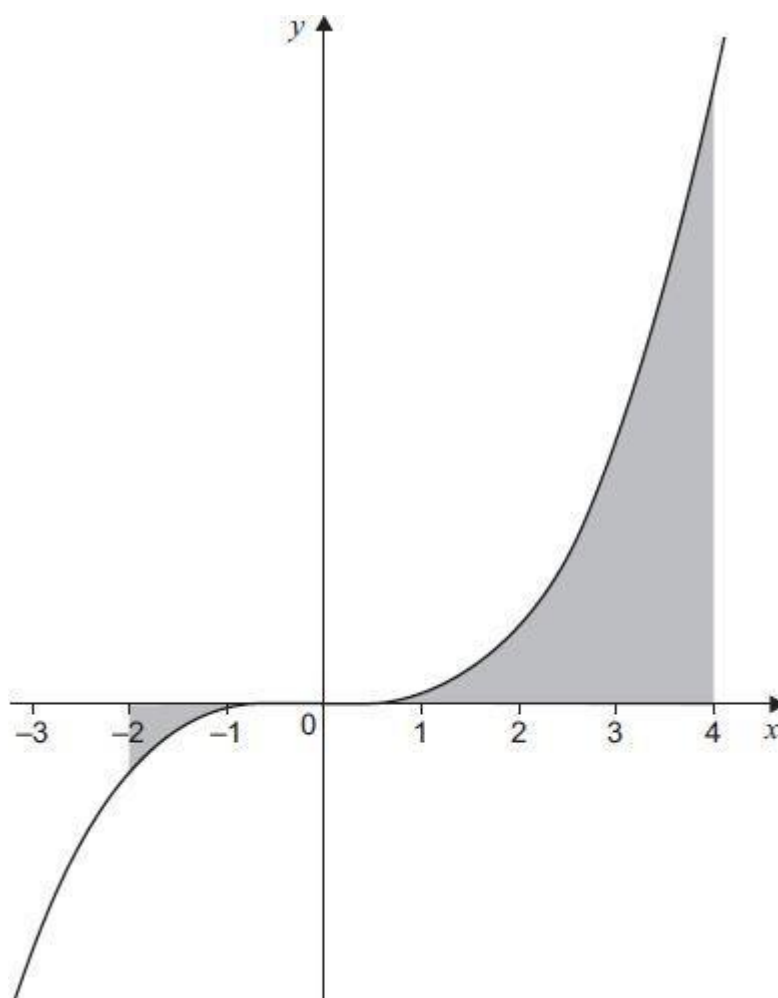
(2)

(Total 12 marks)

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Q2.

The graph of $y = x^3$ is shown.



Find the total shaded area.

Circle your answer.

-68

60

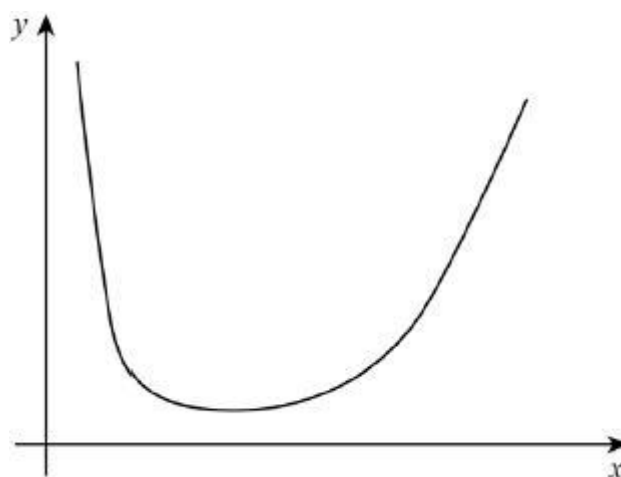
68

128

(Total 1 mark)

Q3.

A curve has equation $y = 6x^2 + \frac{8}{x^2}$ and is sketched below for $x > 0$



Find the area of the region bounded by the curve, the x -axis and the lines $x = a$ and $x = 2a$, where $a > 0$, giving your answer in terms of a

(Total 4 marks)

Q4.

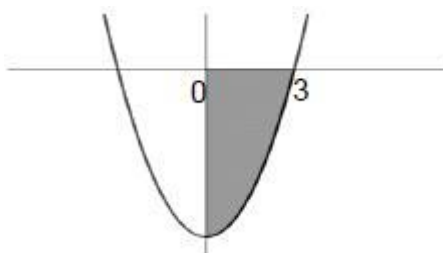
- (a) Given that $u = 2^x$, write down an expression for $\frac{du}{dx}$ (1)

- (b) Find the exact value of $\int_0^1 2^x \sqrt{3 + 2^x} \, dx$
Fully justify your answer. (6)

(Total 7 marks)

Q5.

The graph of $y = x^2 - 9$ is shown below.



Find the area of the shaded region.
Circle your answer.

-18

-6

6

18

(Total 1 mark)

Q6.

$\int_1^2 x^3 \ln(2x) dx$ can be written in the form $p \ln 2 + q$, where p and q are rational numbers.

Find p and q .

(Total 5 marks)

Q7.

Find the value of $\int_1^2 \frac{6x+1}{6x^2-7x+2} dx$, expressing your answer in the form $m \ln 2 + n \ln 3$, where m and n are integers.

(Total 8 marks)

Q8.

- (a) Use the trapezium rule with five ordinates (four strips) to find an approximate value for

$$\int_1^5 \frac{1}{x^2 + 1} dx$$

giving your answer to three significant figures.

(4)

- (b) (i) Find $\int \left(x^{-\frac{3}{2}} + 6x^{\frac{1}{2}} \right) dx$, giving the coefficient of each term in its simplest form.

(3)

- (ii) Hence find the value of $\int_1^4 \left(x^{-\frac{3}{2}} + 6x^{\frac{1}{2}} \right) dx$.

(2)

(Total 9 marks)

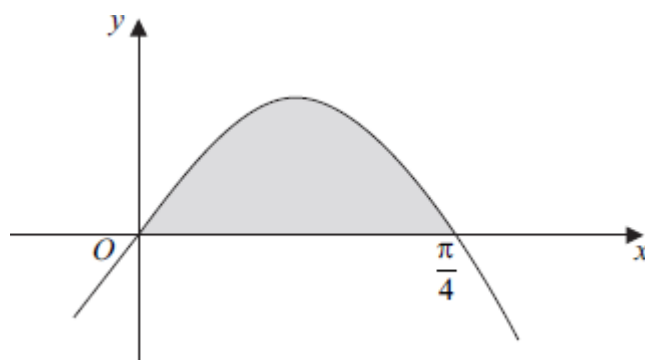
Q9.

A curve has equation $y = 4x \cos 2x$.

- (a) Find an exact equation of the tangent to the curve at the point on the curve where $x = \frac{\pi}{4}$.

(5)

- (b) The region shaded on the diagram below is bounded by the curve $y = 4x \cos 2x$ and the x -axis from $x = 0$ to $x = \frac{\pi}{4}$.



By using integration by parts, find the exact value of the area of the shaded region.

(5)

(Total 10 marks)

Q10.

- (a) Show that

$$\int_0^{\ln 2} e^{1-2x} dx = \frac{3}{8}e$$

(4)

- (b) Use the substitution $u = \tan x$ to find the exact value of

$$\int_0^{\frac{\pi}{4}} \sec^4 x \sqrt{\tan x} dx$$

(8)

(Total 12 marks)

Q11.

A curve has equation $y = x^5 - 2x^2 + 9$. The point P with coordinates $(-1, 6)$ lies on the curve.

- (a) Find the equation of the tangent to the curve at the point P , giving your answer in the form $y = mx + c$.

(5)

- (b) The point Q with coordinates $(2, k)$ lies on the curve.

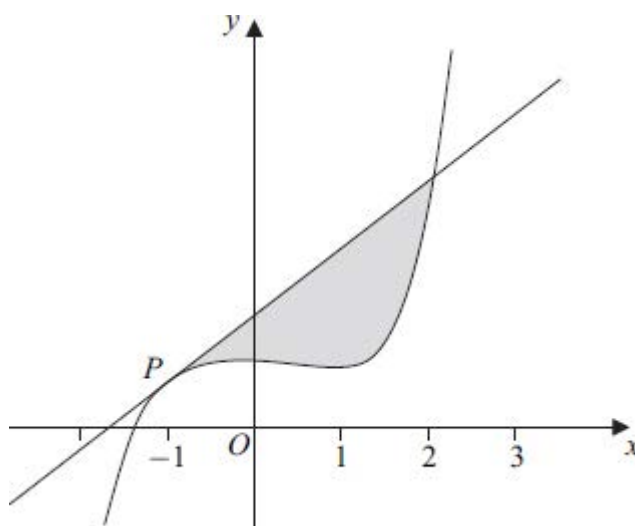
- (i) Find the value of k .

(1)

- (ii) Verify that Q also lies on the tangent to the curve at the point P .

(1)

- (c) The curve and the tangent to the curve at P are sketched below.



- (i) Find $\int_{-1}^2 (x^5 - 2x^2 + 9) dx$. (5)
- (ii) Hence find the area of the shaded region bounded by the curve and the tangent to the curve at P . (3)
- (Total 15 marks)

Q12.

- (a) (i) Using the binomial expansion, or otherwise, express $(2 + y)^3$ in the form $a + by + cy^2 + y^3$, where a , b and c are integers. (2)
- (ii) Hence show that $(2 + x^{-2})^3 + (2 - x^{-2})^3$ can be expressed in the form $p + qx^{-4}$, where p and q are integers. (3)
- (b) (i) Hence find $\int [(2 + x^{-2})^3 + (2 - x^{-2})^3] dx$. (2)
- (ii) Hence find the value of $\int_1^2 [(2 + x^{-2})^3 + (2 - x^{-2})^3] dx$. (2)
- (Total 9 marks)

Q13.

- (a) (i) By writing $\ln x$ as $(\ln x) \times 1$, use integration by parts to find $\int \ln x dx$. (4)

(ii) Find $\int (\ln x)^2 dx$.

(4)

(b) Use the substitution $u = \sqrt{x}$ to find the exact value of $\int_1^4 \frac{1}{x + \sqrt{x}} dx$.

(7)

(Total 15 marks)

Q14.

(a) (i) Express $\frac{5 - 8x}{(2 + x)(1 - 3x)}$ in the form $\frac{A}{2 + x} + \frac{B}{1 - 3x}$, where A and B are integers.

(3)

(ii) Hence show that $\int_{-1}^0 \frac{5 - 8x}{(2 + x)(1 - 3x)} dx = p \ln 2$, where p is rational.

(4)

(b) (i) Given that $\frac{9 - 18x - 6x^2}{2 - 5x - 3x^2}$ can be written as $C + \frac{5 - 8x}{2 - 5x - 3x^2}$, find the value of C .

(1)

(ii) Hence find the exact value of the area of the region bounded by the curve $y = \frac{9 - 18x - 6x^2}{2 - 5x - 3x^2}$, the x -axis and the lines $x = -1$ and $x = 0$.

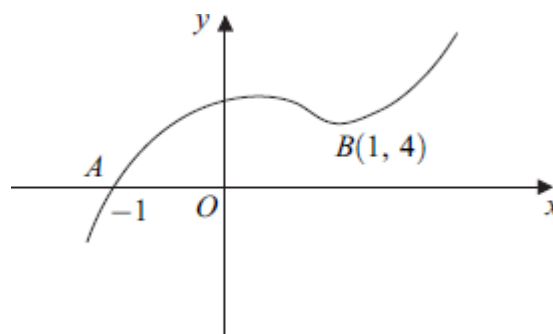
You may assume that $y > 0$ when $-1 \leq x \leq 0$.

(2)

(Total 10 marks)

Q15.

The curve with equation $y = x^5 - 3x^2 + x + 5$ is sketched below. The point O is at the origin and the curve passes through the points $A(-1, 0)$ and $B(1, 4)$.



(a) Given that $y = x^5 - 3x^2 + x + 5$, find:

(i) $\frac{dy}{dx}$;

(3)

(ii) $\frac{d^2y}{dx^2}$.

(1)

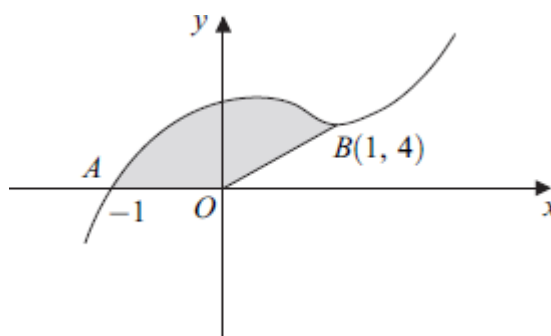
(b) Find an equation of the tangent to the curve at the point $A(-1, 0)$.

(2)

(c) Verify that the point B , where $x = 1$, is a minimum point of the curve.

(3)

(d) The curve with equation $y = x^5 - 3x^2 + x + 5$ is sketched below. The point O is at the origin and the curve passes through the points $A(-1, 0)$ and $B(1, 4)$.



(i) Find $\int_{-1}^1 (x^5 - 3x^2 + x + 5) dx$.

(5)

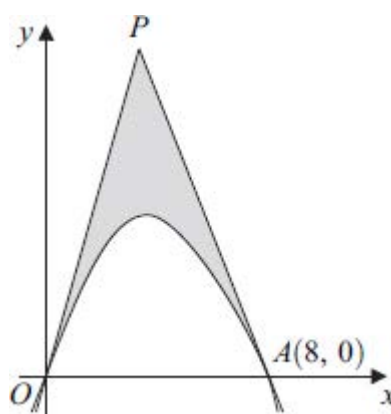
(ii) Hence find the area of the shaded region bounded by the curve between A and B and the line segments AO and OB .

(2)

(Total 16 marks)

Q16.

The diagram shows part of a curve crossing the x -axis at the origin O and at the point $A(8, 0)$. Tangents to the curve at O and A meet at the point P , as shown in the diagram.



The curve has equation

$$y = 12x - 3x^{\frac{5}{3}}$$

- (a) Find $\frac{dy}{dx}$. (2)
- (b) (i) Find the value of $\frac{dy}{dx}$ at the point O and hence write down an equation of the tangent at O . (2)
- (ii) Show that the equation of the tangent at $A(8, 0)$ is $y + 8x = 64$. (3)
- (c) Find $\int \left(12x - 3x^{\frac{5}{3}}\right) dx$. (3)
- (d) Calculate the area of the shaded region bounded by the curve from O to A and the tangents OP and AP . (7)
- (Total 17 marks)**

Q17.

- (a) Given that $y = 4x^3 - 6x + 1$, find $\frac{dy}{dx}$. (1)
- (b) Hence find $\int_2^3 \frac{2x^2 - 1}{4x^3 - 6x + 1} dx$, giving your answer in the form $p \ln q$, where p and q are rational numbers. (5)

(Total 6 marks)

Q18.

- (a) Given that $x = \frac{1}{\sin \theta}$, use the quotient rule to show that $\frac{dx}{d\theta} = -\operatorname{cosec} \theta \cot \theta$. (3)

- (b) Use the substitution $x = \operatorname{cosec} \theta$ to find $\int_{\sqrt{2}}^2 \frac{1}{x^2 \sqrt{x^2 - 1}} dx$, giving your answer to three significant figures. (9)

(Total 12 marks)

Q19.

- (a) Express $\frac{2x+3}{4x^2-1}$ in the form $\frac{A}{2x-1} + \frac{B}{2x+1}$, where A and B are integers. (3)

- (b) Express $\frac{12x^3 - 7x - 6}{4x^2 - 1}$ in the form $Cx + \frac{D(2x+3)}{4x^2 - 1}$, where C and D are integers. (3)

- (c) Evaluate $\int_1^2 \frac{12x^3 - 7x - 6}{4x^2 - 1} dx$, giving your answer in the form $p + \ln q$, where p and q are rational numbers. (5)

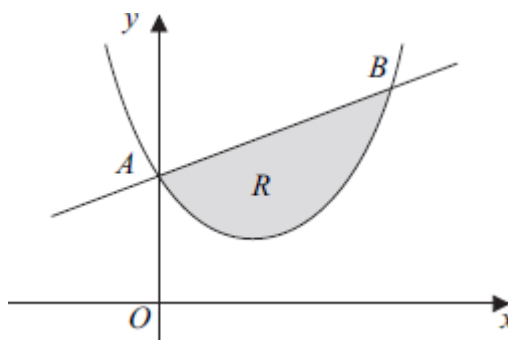
(Total 11 marks)

Q20.

- (a) (i) Express $x^2 - 3x + 5$ in the form $(x - p)^2 + q$. (2)

- (ii) Hence write down the equation of the line of symmetry of the curve with equation $y = x^2 - 3x + 5$. (1)

- (b) The curve C with equation $y = x^2 - 3x + 5$ and the straight line $y = x + 5$ intersect at the point $A(0, 5)$ and at the point B , as shown in the diagram below.



- (i) Find the coordinates of the point B .

(3)

- (ii) Find $\int (x^2 - 3x + 5) dx$.

(3)

- (iii) Find the area of the shaded region R bounded by the curve C and the line segment AB .

(4)

(Total 13 marks)

Q21.

- (a) Expand $\left(x^{\frac{3}{2}} - 1\right)^2$.

(2)

- (b) Hence find $\int \left(x^{\frac{3}{2}} - 1\right)^2 dx$.

(3)

- (c) Hence find the value of $\int_1^4 \left(x^{\frac{3}{2}} - 1\right)^2 dx$.

(2)

(Total 7 marks)

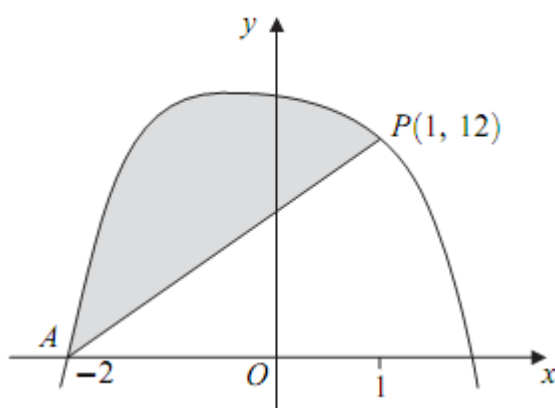
Q22.

Use the substitution $u = x^4 + 2$ to find the value of $\int_0^1 \frac{x^7}{(x^4 + 2)^2} dx$, giving your answer in the form $p \ln q + r$, where p , q and r are rational numbers.

(Total 6 marks)

Q23.

The curve sketched below passes through the point $A (-2, 0)$.

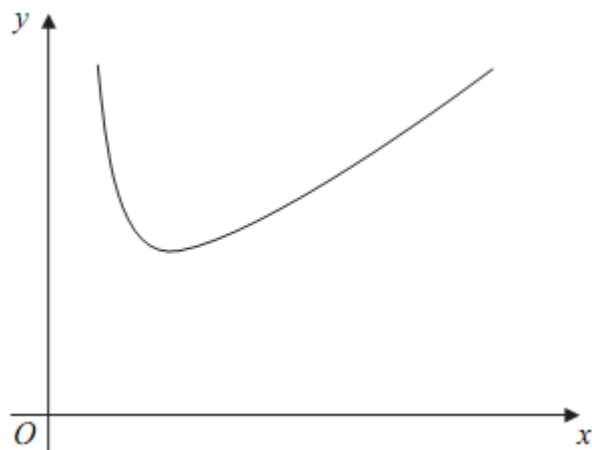


The curve has equation $y = 14 - x - x^4$ and the point $P (1, 12)$ lies on the curve.

- (a) (i) Find the gradient of the curve at the point P . (3)
- (ii) Hence find the equation of the tangent to the curve at the point P , giving your answer in the form $y = mx + c$. (2)
- (b) (i) Find $\int_{-2}^1 (14 - x - x^4) dx$. (5)
- (ii) Hence find the area of the shaded region bounded by the curve $y = 14 - x - x^4$ and the line AP . (2)
- (Total 12 marks)**

Q24.

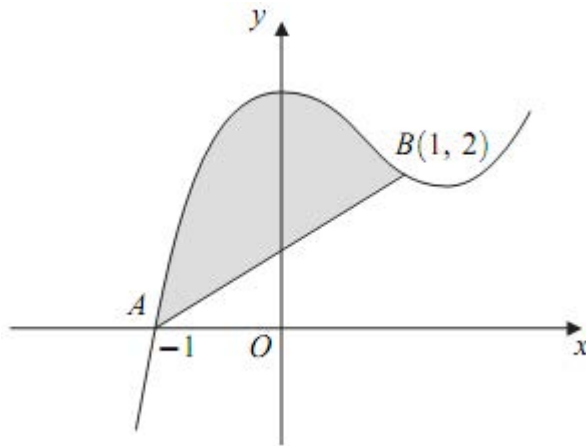
A curve C is defined for $x > 0$ by the equation $y = x + 3 + \frac{8}{x^4}$ and is sketched below.



- (a) Given that $y = x + 3 + \frac{8}{x^4}$, find $\frac{dy}{dx}$. (3)
- (b) Find an equation of the tangent at the point on the curve C where $x = 1$. (3)
- (c) The curve C has a minimum point M . Find the coordinates of M . (4)
- (d) (i) Find $\int \left(x + 3 + \frac{8}{x^4} \right) dx$. (3)
- (ii) Hence find the area of the region bounded by the curve C , the x -axis and the lines $x = 1$ and $x = 2$. (2)
- (e) The curve C is translated by $\begin{bmatrix} 0 \\ k \end{bmatrix}$ to give the curve $y = f(x)$. Given that the x -axis is a tangent to the curve $y = f(x)$, state the value of the constant k . (1)
- (Total 16 marks)**

Q25.

The curve with equation $y = x^3 - 2x^2 + 3$ is sketched below.



The curve cuts the x -axis at the point $A(-1, 0)$ and passes through the point $B(1, 2)$.

- (a) Find $\int_{-1}^1 (x^3 - 2x^2 + 3) dx$. (5)
- (b) Hence find the area of the shaded region bounded by the curve $y = x^3 - 2x^2 + 3$ and the line AB . (3)
- (Total 8 marks)

Q26.

- (a) The expression $(2 + x^2)^3$ can be written in the form

$$8 + px^2 + qx^4 + x^6$$

Show that $p = 12$ and find the value of the integer q .

- (b) (i) Hence find $\int \frac{(2 + x^2)^3}{x^4} dx$. (5)
- (ii) Hence find the exact value of $\int_1^2 \frac{(2 + x^2)^3}{x^4} dx$. (2)
- (Total 10 marks)

Q27.

- (a) A curve is defined by the parametric equations $x = 3 \cos 2\theta$, $y = 2 \cos \theta$.

- (i) Show that $\frac{dy}{dx} = \frac{1}{k \cos \theta}$, where k is an integer.

(4)

- (ii) Find an equation of the normal to the curve at the point where $\theta = \frac{\pi}{3}$.

(4)

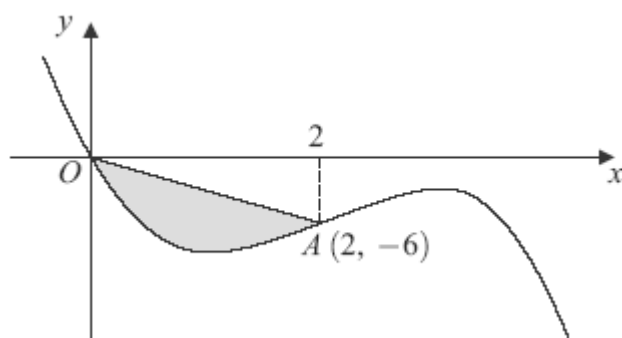
- (b) Find the exact value of $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x dx$.

(5)

(Total 13 marks)

Q28.

The curve with equation $y = 12x^2 - 19x - 2x^3$ is sketched below.



The curve crosses the x -axis at the origin O , and the point $A(2, -6)$ lies on the curve.

- (a) (i) Find the gradient of the curve with equation $y = 12x^2 - 19x - 2x^3$ at the point A .

(4)

- (ii) Hence find the equation of the normal to the curve at the point A , giving your answer in the form $x + py + q = 0$, where p and q are integers.

(3)

- (b) (i) Find the value of $\int_0^2 (12x^2 - 19x - 2x^3) dx$.

(5)

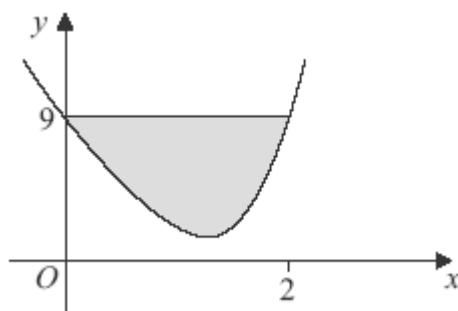
- (ii) Hence determine the area of the shaded region bounded by the curve and the line OA .

(3)

(Total 15 marks)

Q29.

The curve with equation $y = x^4 - 8x + 9$ is sketched below.



The point (2, 9) lies on the curve.

- (a) (i) Find $\int_0^2 (x^4 - 8x + 9) dx$. (5)
- (ii) Hence find the area of the shaded region bounded by the curve and the line $y = 9$. (2)
- (b) The point $A(1, 2)$ lies on the curve with equation $y = x^4 - 8x + 9$.
- (i) Find the gradient of the curve at the point A . (4)
- (ii) Hence find an equation of the tangent to the curve at the point A . (1)
- (Total 12 marks)

Q30.

- (a) The expression $\left(1 - \frac{1}{x^2}\right)^3$ can be written in the form
- $$1 + \frac{p}{x^2} + \frac{q}{x^4} - \frac{1}{x^6}$$

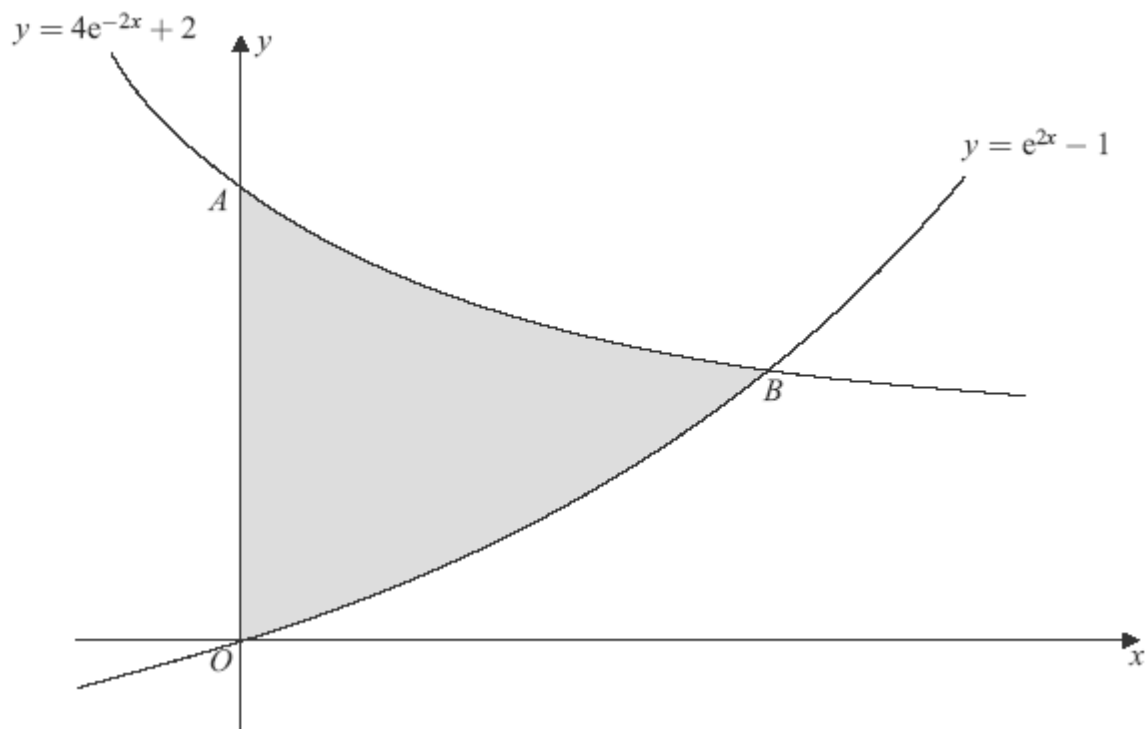
Find the values of the integers p and q .

- (b) (i) Hence find $\int \left(1 - \frac{1}{x^2}\right)^3 dx$. (4)
- (ii) Hence find the value of $\int_{\frac{1}{2}}^1 \left(1 - \frac{1}{x^2}\right)^3 dx$. (2)

(Total 8 marks)

Q31.

The diagram shows the curves $y = e^{2x} - 1$ and $y = 4e^{-2x} + 2$.



The curve $y = 4e^{-2x} + 2$ crosses the y -axis at the point A and the curves intersect at the point B .

- (a) Describe a sequence of two geometrical transformations that maps the graph of $y = e^x$ onto the graph of $y = e^{2x} - 1$.

(4)

- (b) Write down the coordinates of the point A .

(1)

- (c) (i) Show that the x -coordinate of the point B satisfies the equation

$$(e^{2x})^2 - 3e^{2x} - 4 = 0$$

(2)

- (ii) Hence find the exact value of the x -coordinate of the point B .

(3)

- (d) Find the exact value of the area of the shaded region bounded by the curves $y = e^{2x} - 1$ and $y = 4e^{-2x} + 2$ and the y -axis.

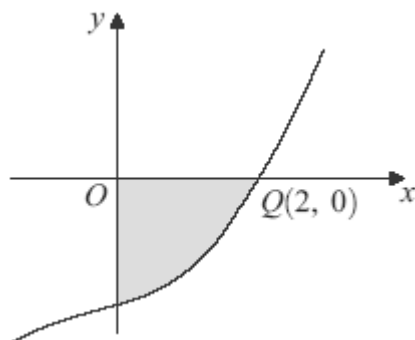
(5)

(Total 15 marks)

Q32.

- (a) The polynomial $p(x)$ is given by $p(x) = x^3 + x - 10$.

- (i) Use the Factor Theorem to show that $x - 2$ is a factor of $p(x)$. (2)
- (ii) Express $p(x)$ in the form $(x - 2)(x^2 + ax + b)$, where a and b are constants. (2)
- (b) The curve C with equation $y = x^3 + x - 10$, sketched below, crosses the x -axis at the point $Q(2, 0)$.



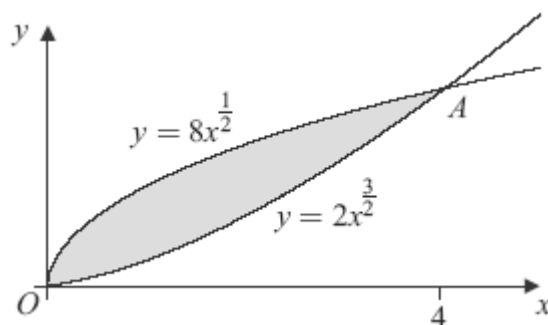
- (i) Find the gradient of the curve C at the point Q . (4)
- (ii) Hence find an equation of the tangent to the curve C at the point Q . (2)
- (iii) Find $\int (x^3 + x - 10) dx$. (3)
- (iv) Hence find the area of the shaded region bounded by the curve C and the coordinate axes. (2)

(Total 15 marks)

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Q33.

The diagram shows a sketch of the curves with equations $y = 2x^{\frac{3}{2}}$ and $y = 8x^{\frac{1}{2}}$.

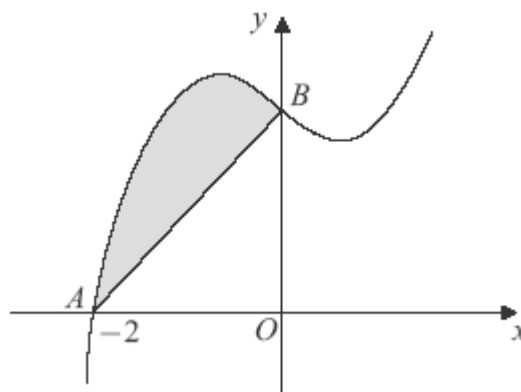


The curves intersect at the origin and at the point A , where $x = 4$.

- (a) (i) For the curve $y = 2x^{\frac{3}{2}}$, find the value of $\frac{dy}{dx}$ when $x = 4$. (2)
- (ii) Find an equation of the normal to the curve $y = 2x^{\frac{3}{2}}$ at the point A . (4)
- (b) (i) Find $\int 8x^{\frac{1}{2}} dx$. (2)
- (ii) Find the area of the shaded region bounded by the two curves. (4)
- (c) Describe a single geometrical transformation that maps the graph of $y = 2x^{\frac{3}{2}}$ onto the graph of $y = 2(x+3)^{\frac{3}{2}}$. (2)
- (Total 14 marks)

Q34.

- (a) The polynomial $p(x)$ is given by $p(x) = x^3 - x + 6$.
- (i) Use the Factor Theorem to show that $x + 2$ is a factor of $p(x)$. (2)
- (ii) Express $p(x) = x^3 - x + 6$ in the form $(x + 2)(x^2 + bx + c)$, where b and c are integers. (2)
- (iii) The equation $p(x) = 0$ has one root equal to -2 . Show that the equation has no other real roots. (2)
- (b) The curve with equation $y = x^3 - x + 6$ is sketched below.



The curve cuts the x -axis at the point $A(-2, 0)$ and the y -axis at the point B .

- (i) State the y -coordinate of the point B .

(1)

- (ii) Find $\int_{-2}^0 (x^3 - x + 6) dx$.

(5)

- (iii) Hence find the area of the shaded region bounded by the curve $y = x^3 - x + 6$ and the line AB .

(3)

(Total 15 marks)

Q35.

- (a) Write down the value of n given that $\frac{1}{x^4} = x^n$.

(1)

- (b) Expand $\left(1 + \frac{3}{x^2}\right)^2$.

(2)

- (c) Hence find $\int \left(1 + \frac{3}{x^2}\right)^2 dx$.

(3)

- (d) Hence find the exact value of $\int_1^3 \left(1 + \frac{3}{x^2}\right)^2 dx$.

(2)

(Total 8 marks)